

Homogenization benchmark of a thin-walled MSG shell model against 3D solid finite elements: isotropic and $[45/-45]$ circular tubes across the thin-to-thick range

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Abstract

We benchmark a thin-walled Mechanics-of-Structure-Genome (MSG) shell cross-sectional analysis—in both Reissner–Mindlin (RM, C^0) and Kirchhoff (C^1) wall kinematics—against a three-dimensional (3D) solid finite-element cross-sectional analysis (FEniCSx), the latter taking the role usually played by VABS. Two circular composite tubes are studied: an isotropic tube and a balanced anisotropic $[45/-45]$ tube. The wall slenderness h/R is swept over five values from thin-shell (0.01) to thick-shell (0.20), and the shell wall is referenced at either the mid-surface (“centre”) or the outer mould line (“OML”). The full Timoshenko 6×6 stiffness is compared term by term and the isotropic case is additionally checked against the exact annulus closed form. *Key results.* At the centre reference, the shell reproduces the solid to $< 1\%$ on every non-zero stiffness (EA , GA , GJ , EI) across the entire thin-to-thick range, for both RM and Kirchhoff. At the OML reference the error grows roughly linearly with h/R : EA and EI over-predict by $\approx h/2R$ (the $(1 + z/R)$ curvature/arc-length factor) and the two shell models diverge on GA and GJ . The $[45/-45]$ tube exhibits the expected high torsional stiffness ($GJ/EA \approx 0.85$), captured to $< 1\%$ at the centre; its extension–twist coupling is near zero (balanced layup). The practical recommendation is to reference the wall at its mid-surface, where the thin-walled shell model is essentially exact through $h/R = 0.20$.

1 Introduction

Cross-sectional homogenization reduces a slender composite structure to a 1D beam with an effective 6×6 stiffness, after which a global beam solution and a dehomogenization (3D stress recovery) complete the analysis. For thin-walled sections, a *shell* idealization of the wall is far cheaper than a 2D *solid* discretization, but its accuracy depends on the wall slenderness h/R and on the choice of wall reference surface. This report quantifies both dependencies by benchmarking an MSG thin-walled shell model (RM and Kirchhoff) against a 3D solid analysis for isotropic and $[45/-45]$ tubes, over a thin-to-thick h/R sweep, at the mid-surface and OML references.

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2 Model and method

2.1 Thin-walled MSG shell (summary)

The wall is a 1D structure genome (the mid-surface curve) whose through-thickness behaviour is condensed into an 8×8 plate law

$$\begin{Bmatrix} N \\ M \\ Q \end{Bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B} & \mathbf{0} \\ \mathbf{B} & \mathbf{D} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{G} \end{bmatrix} \begin{Bmatrix} \varepsilon^0 \\ \kappa \\ \gamma^s \end{Bmatrix}, \quad (1)$$

with the membrane–bending block $[\mathbf{A}, \mathbf{B}; \mathbf{B}, \mathbf{D}]$ from classical lamination theory and the transverse-shear block $\mathbf{G}$ from a first-order shear-deformation (FSDT) 3D-equilibrium argument (reducing to $\frac{5}{6}Gh$ for an isotropic plate); this is an engineering approximation to the asymptotically-exact transverse-shear stiffness, which in the rigorous RM shell theory follows from energy minimization rather than an assumed correction factor [1]. Two wall kinematics are used on the same genome: a Kirchhoff model (no transverse shear, C^1 cubic Hermite) and a Reissner–Mindlin model (transverse shear retained, C^0 Lagrange with selective reduced integration for the shear energy). The Euler–Bernoulli (4×4) and Timoshenko (6×6) stiffnesses follow from the asymptotic fluctuation fields $\mathbf{V}_0$ and $\mathbf{V}_1$. Full kinematics and the strain field are given in the companion paper; here we focus on the benchmark.

2.2 Reference surface

The shell ABD is referenced either to the wall mid-surface (*centre*, $\text{frac} = 0.5$) or to the outer mould line (*OML*, $\text{frac} = 0$). A reference shift by a through-thickness offset z_0 uses the parallel-axis relation $\mathbf{B} \rightarrow \mathbf{B} - z_0 \mathbf{A}$, $\mathbf{D} \rightarrow \mathbf{D} - 2z_0 \mathbf{B} + z_0^2 \mathbf{A}$, and the wall reference curvature is $k_{22} = -1/r_{\text{ref}}$ with $r_{\text{ref}} = R$ (*centre*) or $R + h/2$ (*OML*). The present flat-plate ABD omits the $(1 + z/R)$ shell metric in the through-thickness integration; the consequence is quantified below. For the symmetric tubes studied here the tension and shear centres coincide with the geometric centre, so the centre-referenced 6×6 is itself the reference-invariant comparison [2]; the OML results are a deliberate reference-sensitivity study, not an alternative “stiffness” of the tube.

2.3 Solid reference

The 3D reference is a 2D solid cross-sectional analysis (FEniCSx, first-order quadrilaterals) with the full 3D constitutive law and the lamina orientation in the element frame; it plays the role of VABS. The annulus is meshed with $N_c = 200$ circumferential and $N_r = 16$ radial elements (8 per ply for $[45/-45]$). For the isotropic tube the solid reproduces the exact annulus closed form ($EA = E\pi(r_o^2 - r_i^2)$, $EI = E\frac{\pi}{4}(r_o^4 - r_i^4)$, $GJ = 2EI G/E$) to $< 0.1\%$ at every h/R , confirming mesh convergence.

2.4 Benchmark matrix

3 Results

The per-stiffness tables (errors relative to the solid) are collected in Tables 2–10; the convergence with h/R is plotted in Figs. 1–4.

Table 1: Benchmark cases. $R = 1$ m; Timoshenko order $[EA, GA_2, GA_3, GJ, EI_2, EI_3]$.

isotropic tube	$E = 70$ GPa, $\nu = 0.3$
anisotropic tube	$[45/-45]$, $E_1=37$, $E_2=E_3=9$, $G=4$ GPa, $\nu=0.3$
h/R sweep	0.01, 0.03, 0.06, 0.12, 0.20
references	centre (frac=0.5), OML (frac=0)
shell models	Reissner–Mindlin, Kirchhoff
reference solution	FEniCS 2D-solid (VABS-equivalent)

3.1 Isotropic tube

At the **centre** reference the shell is essentially exact: EA within 0.01%, and GA , GJ , EI within 1% even at $h/R = 0.20$ (Table 2–5, Fig. 1). Both RM and Kirchhoff overlap with the solid and with the analytic annulus values. At the **OML** reference the error grows almost linearly in h/R : EA and EI over-predict by $\approx h/2R$ (+10% and +9.4% at $h/R = 0.20$), consistent with the OML radius being $(1 + h/2R)$ times the mid-surface radius (the omitted curvature metric). GA under-predicts and the two models split (GJ : RM +20%, KF −10% at $h/R = 0.20$).

3.2 Anisotropic $[45/-45]$ tube

The balanced $[45/-45]$ layup yields a high torsional stiffness ($GJ/EA \approx 0.85$), reproduced to $< 1\%$ at the centre reference across all h/R (Table 8, Fig. 3). EA , GA and EI behave as in the isotropic case: $< 1\%$ at the centre, $\sim h/2R$ drift at the OML. The extension–twist coupling C_{14} is two-to-three orders below EA (the +45 and −45 contributions nearly cancel); the large *relative* error there (Table 10) is therefore on a physically negligible term and is not indicative of a modelling deficiency. (A single -45° ply, by contrast, has a significant tension–twist coupling.)

3.3 Reference surface: centre vs OML

The decisive finding is the reference dependence. The mid-surface reference keeps every principal stiffness within 1% of the solid from thin to thick shells; the OML reference incurs an $O(h/R)$ error that reaches 10–20% at $h/R = 0.20$. The over-prediction of EA/EI matches the $(1 + h/2R)$ arc-length factor expected when the wall is integrated about the outer surface without the $(1 + z/R)$ shell metric. For design use the wall should be referenced at its mid-surface.

3.4 Reissner–Mindlin vs Kirchhoff

At the centre reference RM and Kirchhoff are both within 1% and nearly coincide; the transverse-shear kinematics of RM matter most for GA and for the OML reference, where the two models diverge (RM and KF bracket the solid GJ). For the recovered transverse-shear *stress* (not a homogenization quantity) the RM model is essential, since Kirchhoff returns it as identically zero; that behaviour is documented in the companion paper.

4 Discussion

Three points stand out. (i) The thin-walled MSG shell is not merely a thin-limit approximation: at the mid-surface reference it reproduces the 3D solid to $< 1\%$ well into the thick-shell range ($h/R = 0.20$), for isotropic and balanced anisotropic walls alike, in both RM and Kirchhoff forms.

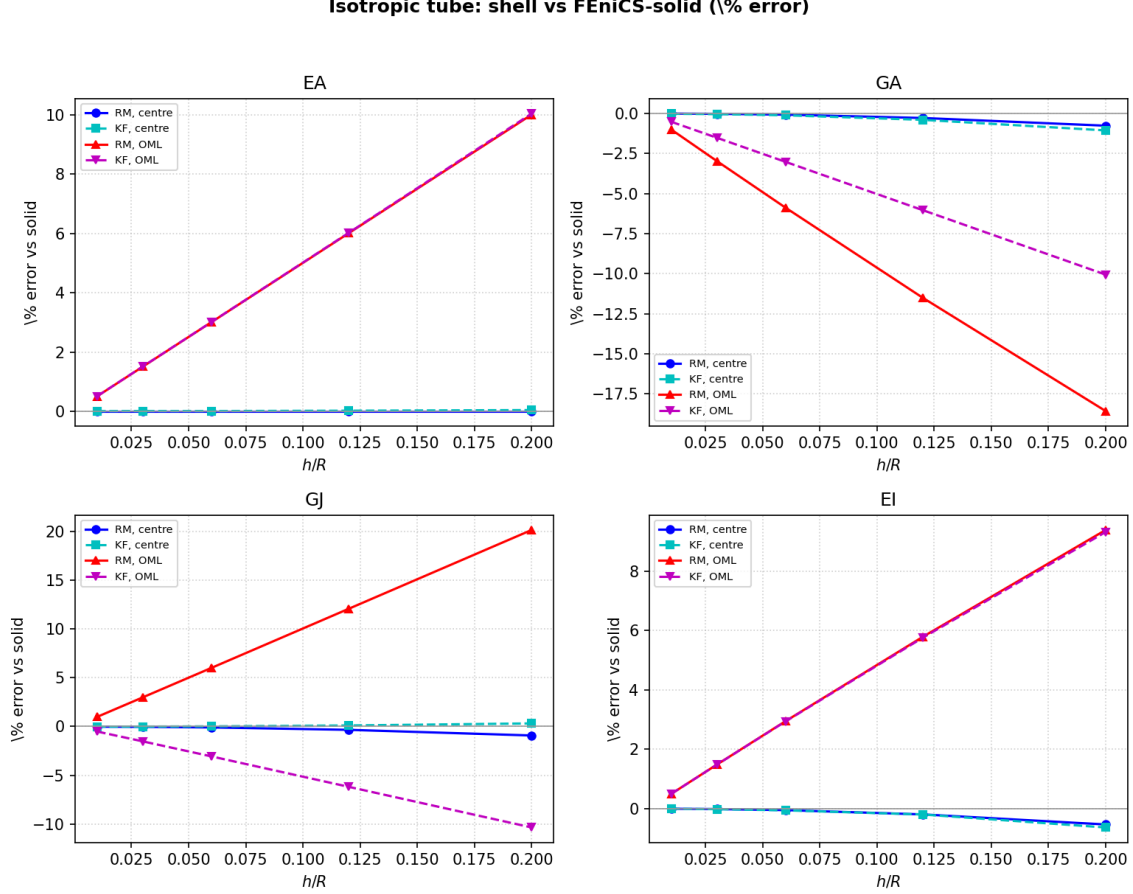


Figure 1: Isotropic tube: shell-vs-solid % error in each stiffness vs h/R . The centre-reference curves stay near zero through the thick-shell range; the OML curves grow $\sim$ linearly and the RM/KF models diverge on GA and GJ .

(ii) The dominant error source for thick walls is the *reference-surface* choice, not the shell kinematics: the OML reference omits the $(1 + z/R)$ curvature metric and incurs an $O(h/R)$ bias, whereas the mid-surface reference is symmetric and the leading curvature term integrates to zero [1]. (iii) RM and Kirchhoff agree at the centre; the model split that appears *only* at the OML reflects the off-centric configuration activating a transverse-shear measure ($2\gamma_{12} \sim \kappa_1 h$) that the Kirchhoff kinematics cannot carry [4], which is why GA and GJ —the shear-sensitive terms—are where the two models diverge. RM’s standing advantage is in transverse-shear quantities (GA and, in dehomogenization, the transverse-shear stress, which Kirchhoff returns as identically zero). The sub-1% centre-reference residual on GA/GJ is this Kirchhoff-vs-RM shear gap (and the FSDT shear approximation), not a missing shear-correction factor. The small extension–twist coupling of the balanced $[45/-45]$ tube ($A_{16} = A_{26} = 0$, the ± 45 contributions cancelling [4]) is a reminder to judge couplings by absolute magnitude, not relative error; a single -45° ply, by contrast, carries an $O(1)$ tension–twist coupling.

The physical interpretations above were cross-checked against the asymptotic shell/beam literature (the curvature Jacobian and $O(h/R)$ energy ordering [1]; the off-centric transverse-shear activation and balanced-layup decoupling [4]; the reference-centre invariants [2]).

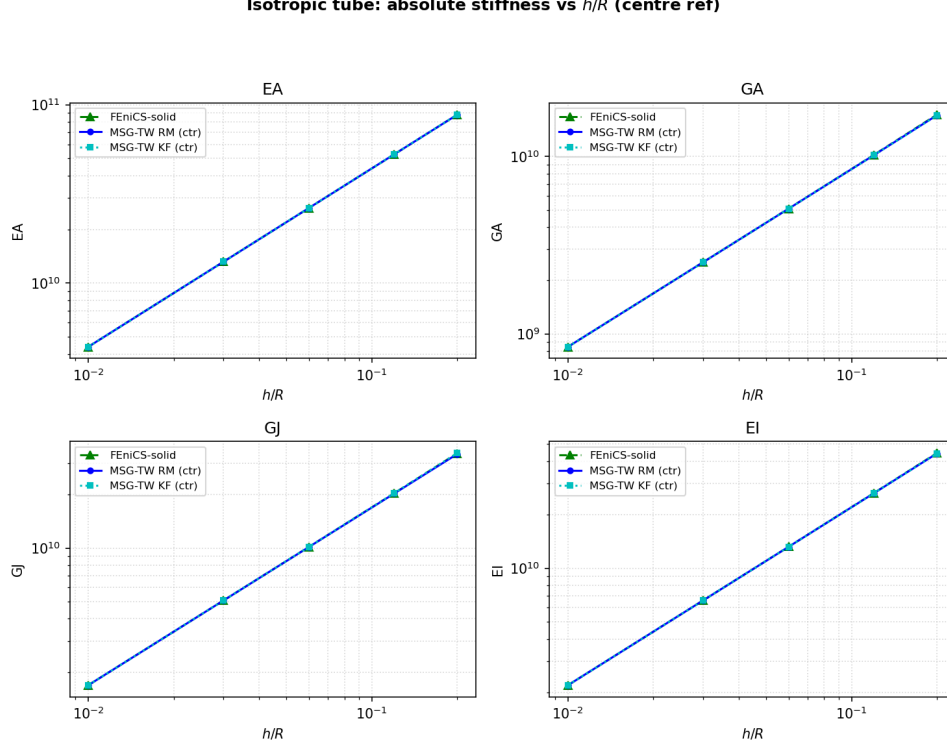


Figure 2: Isotropic tube: absolute stiffness vs h/R (centre reference, log–log). Shell (RM, KF) and solid overlap across the full range.

5 Conclusions

A thin-walled MSG shell cross-sectional analysis, referenced at the wall mid-surface, matches a 3D solid (VABS-equivalent) analysis to better than 1% on the full Timoshenko stiffness of isotropic and [45/–45] circular tubes, across $h/R = 0.01$ – 0.20 , in both Reissner–Mindlin and Kirchhoff forms. The outer-mould-line reference introduces an $O(h/R)$ curvature/arc-length error that reaches 10–20% for thick walls and splits the two shell models; the mid-surface reference is therefore recommended. The high torsional stiffness of the balanced anisotropic layup is captured to $< 1\%$. These results delineate the validity envelope of the thin-walled shell model and confirm it is accurate far beyond the nominally “thin” regime when the wall reference is chosen correctly.

Reproducibility. All cases are generated by `bench_solid.py` (FEniCS 2D-solid, WSL), `bench_shell.py` (JAX MSG-TW shell), and `bench_report.py` (tables and figures); raw 6×6 data are in `data/`.

References

- [1] W. Yu, D. H. Hodges, V. V. Volovoi, Asymptotic generalization of Reissner–Mindlin theory: accurate three-dimensional recovery for composite shells, *Comput. Methods Appl. Mech. Engrg.* 191 (2002) 5087–5109.
- [2] W. Yu, D. H. Hodges, J. C. Ho, Variational asymptotic beam sectional analysis — an updated version, *Int. J. Eng. Sci.* 59 (2012) 40–64.

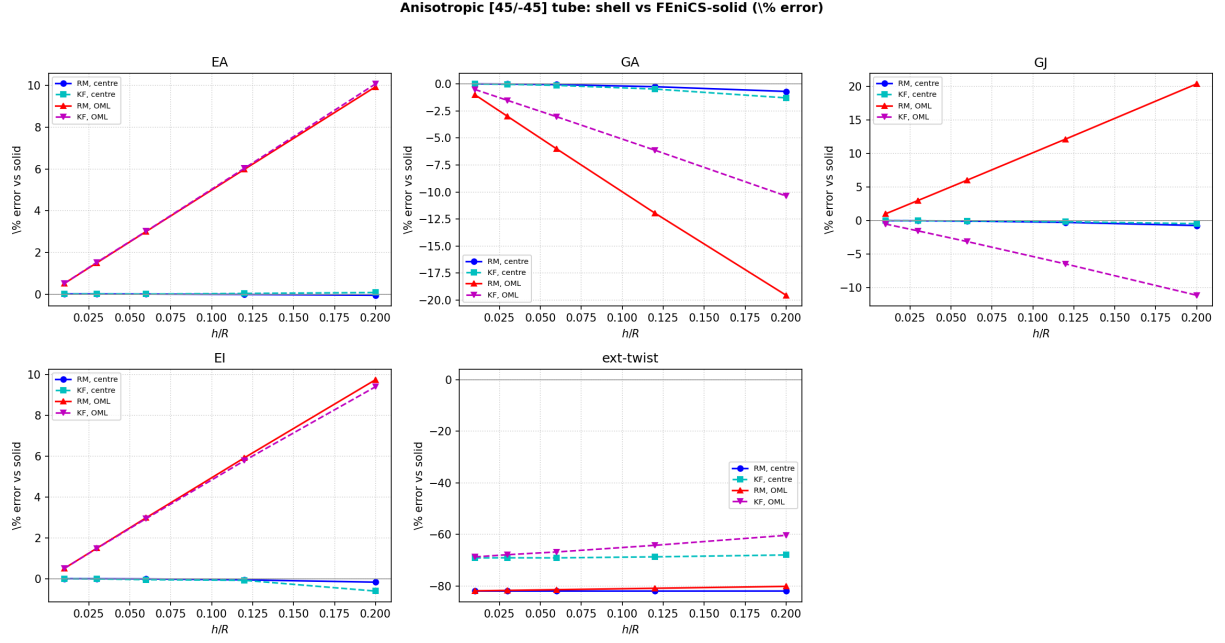


Figure 3: Anisotropic [45/-45] tube: shell-vs-solid % error vs h/R . The principal stiffnesses mirror the isotropic case; the ext-twist panel is the near-zero balanced-layup coupling (large relative error on a negligible term).

Table 2: Isotropic tube: EA vs h/R (shell vs FEniCS-solid; % relative to solid).

h/R	solid	RM-ctr%	KF-ctr%	RM-OML%	KF-OML%	analytic%
0.01	4.3975e+09	+0.01	+0.01	+0.51	+0.51	+0.02
0.03	1.3193e+10	+0.01	+0.01	+1.51	+1.52	+0.02
0.06	2.6385e+10	+0.01	+0.01	+3.01	+3.01	+0.02
0.12	5.2770e+10	+0.01	+0.02	+6.01	+6.02	+0.02
0.20	8.7950e+10	+0.01	+0.04	+10.01	+10.05	+0.02

- [3] A. Deo, W. Yu, Thin-walled composite beam cross-sectional analysis using the mechanics of structure genome, AIAA SciTech, 2020.
- [4] A. Bagla et al., Theory of arbitrarily-shaped thin-walled composite beams via the mechanics of structure genome, Proc. American Society for Composites (ASC), 2025.

Anisotropic [45/-45] tube: absolute stiffness vs h/R (centre ref)

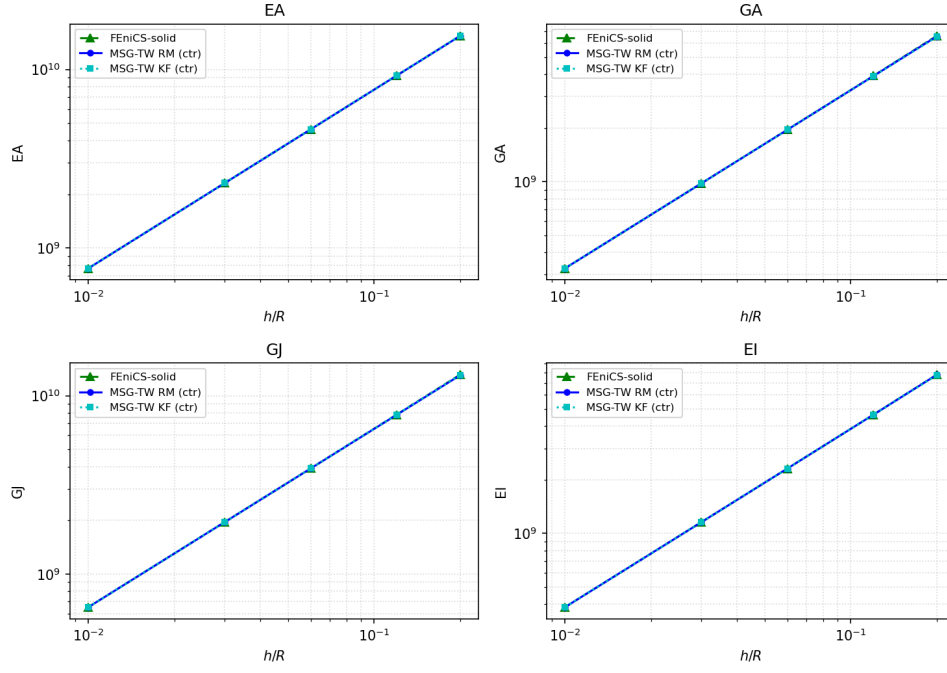


Figure 4: Anisotropic [45/-45] tube: absolute stiffness vs h/R (centre reference). Note the high GJ ($\approx 0.85 EA$).

Table 3: Isotropic tube: GA vs h/R (shell vs FEniCS-solid; % relative to solid).

h/R	solid	RM-ctr%	KF-ctr%	RM-OML%	KF-OML%	analytic%
0.01	8.4578e+08	-0.00	-0.01	-1.00	-0.51	+0.00
0.03	2.5382e+09	-0.02	-0.03	-2.97	-1.51	-0.03
0.06	5.0820e+09	-0.07	-0.11	-5.88	-3.02	-0.14
0.12	1.0209e+10	-0.28	-0.39	-11.51	-6.03	-0.58
0.20	1.7193e+10	-0.76	-1.05	-18.59	-10.06	-1.61

Table 4: Isotropic tube: GJ vs h/R (shell vs FEniCS-solid; % relative to solid).

h/R	solid	RM-ctr%	KF-ctr%	RM-OML%	KF-OML%	analytic%
0.01	1.6911e+09	-0.01	-0.01	+0.99	-0.51	+0.03
0.03	5.0743e+09	-0.03	-0.00	+2.99	-1.53	+0.03
0.06	1.0156e+10	-0.09	+0.02	+6.01	-3.06	+0.03
0.12	2.0366e+10	-0.34	+0.11	+12.07	-6.18	+0.03
0.20	3.4160e+10	-0.92	+0.32	+20.15	-10.35	+0.03

Table 5: Isotropic tube: EI vs h/R (shell vs FEniCS-solid; % relative to solid).

h/R	solid	RM-ctr%	KF-ctr%	RM-OML%	KF-OML%	analytic%
0.01	2.1985e+09	-0.00	-0.00	+0.50	+0.50	+0.03
0.03	6.5967e+09	-0.01	-0.01	+1.49	+1.49	+0.03
0.06	1.3202e+10	-0.05	-0.06	+2.95	+2.94	+0.03
0.12	2.6476e+10	-0.19	-0.19	+5.79	+5.76	+0.03
0.20	4.4408e+10	-0.53	-0.63	+9.39	+9.32	+0.03

Table 6: Anisotropic [45/-45] tube: EA vs h/R (shell vs FEniCS-solid; % relative to solid).

h/R	solid	RM-ctr%	KF-ctr%	RM-OML%	KF-OML%
0.01	7.7054e+08	+0.01	+0.01	+0.51	+0.51
0.03	2.3117e+09	+0.01	+0.01	+1.51	+1.53
0.06	4.6235e+09	+0.00	+0.00	+3.00	+3.02
0.12	9.2485e+09	-0.01	+0.04	+5.99	+6.04
0.20	1.5420e+10	-0.05	+0.08	+9.94	+10.08

Table 7: Anisotropic [45/-45] tube: GA vs h/R (shell vs FEniCS-solid; % relative to solid).

h/R	solid	RM-ctr%	KF-ctr%	RM-OML%	KF-OML%
0.01	3.2597e+08	+0.00	-0.01	-1.00	-0.51
0.03	9.7812e+08	-0.01	-0.04	-3.00	-1.52
0.06	1.9577e+09	-0.06	-0.13	-6.02	-3.04
0.12	3.9268e+09	-0.25	-0.47	-11.96	-6.14
0.20	6.5890e+09	-0.69	-1.29	-19.56	-10.37

Table 8: Anisotropic [45/-45] tube: GJ vs h/R (shell vs FEniCS-solid; % relative to solid).

h/R	solid	RM-ctr%	KF-ctr%	RM-OML%	KF-OML%
0.01	6.5178e+08	-0.01	-0.01	+0.99	-0.51
0.03	1.9557e+09	-0.02	-0.02	+3.00	-1.54
0.06	3.9134e+09	-0.07	-0.05	+6.03	-3.13
0.12	7.8437e+09	-0.27	-0.17	+12.15	-6.47
0.20	1.3140e+10	-0.73	-0.46	+20.37	-11.13

Table 9: Anisotropic [45/-45] tube: EI vs h/R (shell vs FEniCS-solid; % relative to solid).

h/R	solid	RM-ctr%	KF-ctr%	RM-OML%	KF-OML%
0.01	3.8523e+08	+0.00	+0.00	+0.50	+0.50
0.03	1.1559e+09	-0.00	-0.01	+1.50	+1.49
0.06	2.3136e+09	-0.01	-0.05	+2.99	+2.95
0.12	4.6405e+09	-0.06	-0.08	+5.93	+5.78
0.20	7.7873e+09	-0.17	-0.61	+9.75	+9.40

Table 10: Anisotropic [45/-45] tube: ext-twist vs h/R (shell vs FEniCS-solid; % relative to solid).

h/R	solid	RM-ctr%	KF-ctr%	RM-OML%	KF-OML%
0.01	1.4575e+06	-82.00	-69.12	-81.91	-68.75
0.03	1.3118e+07	-82.00	-69.09	-81.73	-67.88
0.06	5.2466e+07	-81.99	-69.11	-81.45	-66.82
0.12	2.0980e+08	-81.99	-68.71	-80.91	-64.23
0.20	5.8239e+08	-81.98	-67.99	-80.17	-60.37